



Automorphism group of a graph related to zero-divisor graphs

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ARTICLE INFO

Article history:

Received 5 December 2023

Received in revised form 13 December 2024

Accepted 8 January 2025

Available online 18 January 2025

Keywords:

Zero-divisor graphs

Automorphisms of graphs

Clique number

ABSTRACT

In 2016, the authors of Wang (2016) have determined the automorphisms of the zero-divisor graph $\Gamma(M_n(\mathbb{F}_q))$, where the graph is defined in such a way: the vertices are all nonzero zero-divisors of $M_n(\mathbb{F}_q)$ and there is a directed edge from A to B if and only if $AB = 0$. Regrettably, the graph $\Gamma(M_n(\mathbb{F}_q))$ is over square matrices, directed, finite and with some loops. Thus we are motivated to improve the definition such that the new defined graph can be infinite, undirected, without loops and it can be over rectangular matrices. Let $M_{m,n}(\mathbb{F})$ be the set of all $m \times n$ rectangular matrices over the field of complex (or real) number. By A^T we mean the conjugate transpose of $A \in M_{m,n}(\mathbb{F})$. Let $\Omega(M_{m,n}(\mathbb{F}))$ be the graph whose vertex set consists of all nonzero matrices in $M_{m,n}(\mathbb{F})$ with rank less than n , and two vertices A, B are adjacent if and only if $AB^T = 0$ (if $\mathbb{F}$ is the real field, that is $AB^T = 0$). Note that $AB^T = 0$ if and only if $BA^T = 0$ and $AA^T \neq 0$ for a nonzero matrix $A \in M_{m,n}(\mathbb{F})$. Thus $\Omega(M_{m,n}(\mathbb{F}))$ is infinite, undirected and without loops. In the present paper, the automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$ are determined completely.

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1. Introduction

The graphs considered in this paper are related to zero-divisor graphs over rings (or over semigroups with zero). Let R be a commutative ring with identity 1 and $M_{m,n}(R)$ the set of all $m \times n$ matrices over R ($M_{n,n}(R)$ is simplified to $M_n(R)$). The zero-divisor graph of R , denoted by $\Gamma(R)$, is an undirected graph with vertices $Z^*(R)$, the set of nonzero zero-divisors of R , and for distinct elements $x, y \in Z^*(R)$, there is an edge joining x and y if and only if $xy = 0$. The concept of zero-divisor graph was first defined and studied for commutative rings by Beck in [7], and further studied by many authors (see, e.g., Akbari et al. [1], Anderson et al. [2–5], Axtell et al. [6], Joshi et al. [12], Levy et al. [13], Lu et al. [16,17], Lucas [18]; Wu [25,26]). A lot of results about the diameter, the girth of $\Gamma(R)$ and so on have been obtained. The main ideal of these researches is to study the interplay between the ring-theoretic properties of a commutative ring R and the graph-theoretic properties of $\Gamma(R)$. Redmond [20] extended the concept of zero-divisor graphs of commutative rings to zero-divisor graphs of noncommutative rings. The zero-divisor graph $\Gamma(R)$ over a noncommutative ring R is a graph with vertex set $Z^*(R)$ and there is a directed edge from a vertex x to a distinct vertex y if and only if $xy = 0$.

We here only introduce some known results on zero-divisor graphs of matrix rings. Akbari and Mohammadian [1] studied the problem of determining when the zero-divisor graphs of rings are isomorphic, given that the zero-divisor graphs of their matrix rings are isomorphic. Božić and Petrović [8] investigated the properties of (directed) zero-divisor graphs of matrix rings, studied the relation between the diameter of the zero-divisor graph of a commutative ring R and

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¹ Supported by the National Natural Science Foundation of China(No. 12371025)

² Supported by the National Natural Science Foundation of China (No. 12101354)

that of the matrix ring $M_n(R)$. Li [14] proved that when studying $\Gamma(R)$, one can replace R with its total quotient ring, and he determined the girth of $\Gamma(\mathcal{T})$ with $\mathcal{T}$ the triangular matrices over a commutative ring. The diameter, the girth and the bounds for the number of edges of $\Gamma(\mathcal{T})$ were given in [15].

Researches on automorphisms of some graphs over matrices dates back to the 1940s, when Hua [10,11] determined the automorphisms of certain Cayley graphs with vertex set of rectangular matrices or symmetric matrices and two matrices A, B are adjacent if and only if $rk(A - B) = 1$. The topic of the present article are motivated by some recent publications on automorphisms of zero-divisor graphs over rings. In [3], Anderson and Livingston proved that $\text{Aut}(\Gamma(\mathbb{Z}_n))$ is a direct product of symmetric groups for $n \geq 4$ a nonprime integer. For noncommutative cases, it was shown in [9] that $\text{Aut}(\Gamma(R))$ is isomorphic to the symmetric group of degree $p + 1$, when $R = M_2(\mathbb{Z}_p)$ with p a prime. Park and Han [19] proved that $\text{Aut}(\Gamma(R)) \cong S_{q+1}$ for $R = M_2(\mathbb{F}_q)$ with $\mathbb{F}_q$ an arbitrary finite field. In [24], Wong et al. determined the automorphisms of the zero-divisor graph (with loops at some vertices) with vertex set of all rank one upper triangular matrices over a finite field. Wang [21] determined the automorphisms of the zero-divisor graph of the full matrix ring over a finite field. The case when R consists of all upper triangular matrices were studied in [22].

However, it is somewhat regretful that the zero-divisor graphs considered in [9,19,21–24] are all over square matrices, they are all finite directed graphs and sometimes with loops. Thus we are encouraged to improve the definition such that the new defined graph can be infinite, undirected, without loops and it can over rectangular matrices. Let $M_{m,n}(\mathbb{F})$ be the set of all $m \times n$ rectangular matrices with $\mathbb{F}$ the complex field $\mathbb{C}$ or the real field $\mathbb{R}$, where $m \geq 1$ and $n \geq 2$. By $\bar{A}^T$ we mean the conjugate transpose of $A \in M_{m,n}(\mathbb{F})$. If $\mathbb{F} = \mathbb{R}$, then $\bar{A}$ is A itself. Let $\Omega(M_{m,n}(\mathbb{F}))$ be the graph whose vertex set consists of all nonzero matrices in $M_{m,n}(\mathbb{F})$ with rank less than n , and two vertices A, B are adjacent if and only if $A\bar{B}^T = 0$. Note that $A\bar{B}^T = 0$ if and only if $BA^T = 0$ and $AA^T \neq 0$ for a nonzero matrix $A \in M_{m,n}(\mathbb{F})$. Thus $\Omega(M_{m,n}(\mathbb{F}))$ is infinite, undirected and without loops. In the present paper, the automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$ are determined completely. The following two theorems are the main results.

Theorem 1.1. *Let $n \geq 3$ be an positive integer. A bijective mapping φ on the vertex set of $\Omega(M_{m,n}(\mathbb{C}))$ is an automorphism if and only if*

$$\varphi(A) = X\bar{A}U, \text{ or } \varphi(A) = XAU,$$

where $X \in M_m(\mathbb{C})$ is invertible and it depends on A , and $U \in M_n(\mathbb{C})$ is a unitary matrix satisfying $U\bar{U}^T = I$.

When the underlying field is the real field $\mathbb{R}$, two vertices A, B of $\Omega(M_{m,n}(\mathbb{R}))$ are adjacent if and only if $AB^T = 0$.

Theorem 1.2. *Let $n \geq 3$ be an positive integer. A bijective mapping φ on the vertex set of $\Omega(M_{m,n}(\mathbb{R}))$ is an automorphism if and only if*

$$\varphi(A) = XAU,$$

where $X \in M_m(\mathbb{R})$ is invertible and it depends on A , and $U \in M_n(\mathbb{R})$ is an orthogonal matrix satisfying $UU^T = I$.

Remark 1.3. Comparing with [9,19,21,22,24], the originality of this article is as follows:

- The automorphisms of $\Omega(M_{m,n}(\mathbb{R}))$ differ from those of $\Gamma(M_n(\mathbb{F}_q))$. Indeed, multiplying an invertible matrix $P \in GL_m(\mathbb{F})$ from left to any given vertex A is an automorphism of $\Omega(M_{m,n}(\mathbb{R}))$, however, the zero-divisor graph $\Gamma(M_n(\mathbb{F}_q))$ has no automorphism of this type. On the other hand, $\Gamma(M_n(\mathbb{F}_q))$ has an automorphism $A \mapsto PAP^{-1}$ for all vertices A and with $P \in GL_n(\mathbb{F}_q)$, however, $\Omega(M_{m,n}(\mathbb{R}))$ has no automorphism of this type.
- The method used in the present article differs from that for determining the automorphisms of $\Gamma(M_n(\mathbb{F}_q))$. We mainly apply the action of the automorphism group on the maximum cliques to determine the undetermined automorphisms, however, the authors of [9,19,21,22,24] are based on the degrees of vertices to determine the undetermined automorphisms.

We organize this article as follows. In Section 2, we study the maximum clique and the clique number of $\Omega(M_{m,n}(\mathbb{F}))$. By applying the main results obtained in Section 2, we give a proof for Theorems 1.1 and 1.2 in Section 3. The automorphisms of $\Omega(M_{m,2}(\mathbb{F}))$ are considered at the end of this article.

2. Maximum cliques and clique number of $\Omega(M_{m,n}(\mathbb{F}))$

Throughout, we assume $m \geq 1, n \geq 2$, $\mathbb{F}$ is the complex field or the real field. By $\mathbb{F}^n$ we denote the inner-product space consisting of all row vector and with the inner product defined as $(\alpha, \beta) = \alpha\bar{\beta}^T$ for $\alpha, \beta \in \mathbb{F}^n$. If $(\alpha, \beta) = 0$ we say α and β are *orthogonal*. A subset of $\mathbb{F}^n$, without the zero vector, is called an *orthogonal set* if any two distinct vector in this set are orthogonal. A basis with n vectors is called an *orthogonal basis* of $\mathbb{F}^n$ if it is also an orthogonal set. By G we denote the set of all $m \times m$ invertible matrices over $\mathbb{F}$. For $A, B \in M_{m,n}(\mathbb{F})$, if there is $P \in G$ such that $B = PA$, we say B and A are *row-equivalent* and write it as $B \stackrel{G}{=} A$. Let Γ be a simple graph. We write $x \sim y$ to denote x, y are adjacent in Γ . An induced subgraph K of Γ is called a *clique* if any two distinct vertices in K are adjacent. A clique is *maximum* if

there is no clique of Γ with larger cardinality. We denote by $\omega(\Gamma)$ the cardinality of a maximum clique, which is called the *clique number* of Γ .

Making clear maximum cliques of $\Omega(M_{m,n}(\mathbb{F}))$ is helpful for determining the automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$. When we study the relation between the maximum cliques of $\Omega(M_{m,n}(\mathbb{F}))$ we need to introduce three types of automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$.

- Let U be a matrix in $M_n(\mathbb{F})$ satisfying $U\bar{U}^T = I$. If $\mathbb{F}$ is the complex field, U is called a *unitary matrix* and if $\mathbb{F}$ is the real field, it is called an *orthogonal matrix*. Define ϕ_U on the vertex set of $\Omega(M_{m,n}(\mathbb{F}))$ as $\phi_U(A) = AU$. It is a bijective mapping on the vertex set of $\Omega(M_{m,n}(\mathbb{C}))$, and $A\bar{B}^T = 0$ if and only if $(AU)(\bar{B}U)^T = 0$. Thus ϕ_U is an automorphism of $\Omega(M_{m,n}(\mathbb{C}))$.

- Let ρ be a bijective mapping on the vertex set of $\Omega(M_{m,n}(\mathbb{F}))$ such that $\rho(A) = X_A A$ for any vertex A of the graph, where $X_A \in G$ depends on A . Since $A\bar{B}^T = 0$ if and only if $(X_A A)(\bar{X}_B B)^T = 0$, ρ is an automorphism of $\Omega(M_{m,n}(\mathbb{F}))$.

- Let f be a field automorphism of $\mathbb{F}$. Indeed, if $\mathbb{F} = \mathbb{R}$ it is the identity automorphism; if $\mathbb{F} = \mathbb{C}$, then $f(a) = a$ or $f(a) = \bar{a}$ for any $a \in \mathbb{C}$. Extending f to matrix $A = [a_{ij}]_{m \times n}$ as $f(A) = [f(a_{ij})]_{m \times n}$ (the extension of f is also written as f). It is easy to check that the mapping $f : A \mapsto f(A)$ on the vertex set of $\Omega(M_{m,n}(\mathbb{F}))$ is an automorphism of $\Omega(M_{m,n}(\mathbb{F}))$.

Let $E_{ij} \in M_{m,n}(\mathbb{F})$ be the matrix unit which has 1 at the (i, j) -position and has 0 elsewhere, and let $e_i \in \mathbb{F}^n$ be the vector which has 1 at the i th position and has 0 elsewhere. The following lemma gives the clique number of $\Omega(M_{m,n}(\mathbb{F}))$.

Lemma 2.1. *The set $\{E_{1,1}, E_{1,2}, \dots, E_{1,n}\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$ and the clique number of $\Omega(M_{m,n}(\mathbb{F}))$ is n .*

Proof. Clearly $E_{1,i} \sim E_{1,j}$ for $i \neq j$. Thus, $\{E_{1,1}, E_{1,2}, \dots, E_{1,n}\}$ induces a clique and the clique number of $\Omega(M_{m,n}(\mathbb{F}))$ is at least n . Conversely, suppose that $\{A_1, A_2, \dots, A_s\}$ induces a clique of $\Omega(M_{m,n}(\mathbb{F}))$ and suppose α_i is a nonzero row of A_i . From $A_i \bar{A}_j^T = 0$ with $i \neq j$, it follows that $\alpha_i \bar{\alpha}_j^T = 0$. Thus $\{\alpha_1, \alpha_2, \dots, \alpha_s\}$ forms an orthogonal set of $\mathbb{F}^n$ and we have $s \leq n$. Hence, the clique number of $\Omega(M_{m,n}(\mathbb{F}))$ is n and $\{E_{1,1}, E_{1,2}, \dots, E_{1,n}\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$. $\square$

Indeed, an arbitrary maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$ can be made clear.

Lemma 2.2. *A set $\{A_1, A_2, \dots, A_n\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$ if and only if each A_i can be decomposed into the product of a nonzero column vector β_i^T and a nonzero row vector α_i as $A_i = \beta_i^T \alpha_i$, and the set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ forms an orthogonal basis of $\mathbb{F}^n$.*

Proof. If $A_i = \beta_i^T \alpha_i$ for $i = 1, 2, \dots, n$ and $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is an orthogonal basis of $\mathbb{F}^n$, then for $i \neq j$,

$$A_i \bar{A}_j^T = \beta_i^T (\alpha_i \bar{\alpha}_j^T) \bar{\beta}_j = 0,$$

which implies that $\{A_1, A_2, \dots, A_n\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$.

Conversely, suppose the set $\{A_1, A_2, \dots, A_n\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$ and let α_i be a nonzero row vector of A_i . Then from $A_i \bar{A}_j^T = 0$ it follows that $\alpha_i \bar{\alpha}_j^T = 0$ for $i \neq j$. Thus $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is an orthogonal basis of $\mathbb{F}^n$. Let γ be a row vector of A_i , then it is a linear combination of $\alpha_1, \dots, \alpha_n$. Suppose $\gamma = k_1 \alpha_1 + \dots + k_n \alpha_n$. The condition $A_i \bar{A}_j^T = 0$ with $i \neq j$ also leads to $\gamma \bar{\alpha}_j^T = 0$. Considering the inner product of γ with α_j for $j \neq i$, we have $k_j \alpha_j \bar{\alpha}_j^T = 0$. As $\alpha_j \bar{\alpha}_j^T \neq 0$, we have $k_j = 0$ for $j \neq i$. Hence, $\gamma = k_i \alpha_i$. Since each row of A_i is a scalar multiple of α_i we can decompose A_i as $\beta_i^T \alpha_i$, with β_i a nonzero row vector. $\square$

Lemma 2.2 implies that a vertex of a maximum clique must be a rank one matrix. Indeed, the maximum cliques of $\Omega(M_{m,n}(\mathbb{F}))$ cover all rank one matrices.

Lemma 2.3. *If A is a rank one matrix, then it lies in a maximum clique.*

Proof. As the rank of A is 1, it can be decomposed as $A = \beta^T \alpha$, where α, β are nonzero vectors. The nonzero vector α can be extended to an orthogonal basis of $\mathbb{F}^n$ as $\{\alpha, \alpha_2, \dots, \alpha_n\}$. Let $A_j = e_1^T \alpha_j$, for $j = 2, \dots, n$. Then $\{A, A_2, \dots, A_n\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$ (by Lemma 2.2). $\square$

Given two maximum cliques, there is an automorphism sends one of which to the other one.

Lemma 2.4. *Suppose $\mathcal{A}, \mathcal{B}$ are maximum cliques of $\Omega(M_{m,n}(\mathbb{F}))$ with vertex set*

$$V(\mathcal{A}) = \{A_1, A_2, \dots, A_n\}, \quad V(\mathcal{B}) = \{B_1, B_2, \dots, B_n\}.$$

There is a unitary matrix Q (or orthogonal matrix for $\mathbb{F} = \mathbb{R}$) such that $\phi_Q(A_j) \stackrel{G}{=} B_j$ for $j = 1, 2, \dots, n$, and there is an automorphism σ such that $\sigma(A_i) = B_i$ for $i = 1, 2, \dots, n$.

Proof. Following from Lemma 2.2, we may suppose $A_i = \gamma_i^T \alpha_i$ and $B_i = \lambda_i^T \beta_i$ for $i = 1, 2, \dots, n$, where $\gamma_i, \alpha_i, \lambda_i, \beta_i$ are all nonzero vectors; $\{\alpha_1, \dots, \alpha_n\}$ and $\{\beta_1, \dots, \beta_n\}$ both are orthogonal base of $\mathbb{F}^n$. Let $\alpha'_i = \frac{1}{\sqrt{\alpha_i \bar{\alpha}_i^T}} \alpha_i$ and $\beta'_i = \frac{1}{\sqrt{\beta_i \bar{\beta}_i^T}} \beta_i$ for

$1 \leq i \leq n$. There is a unitary matrix Q (or orthogonal matrix for $\mathbb{F} = \mathbb{R}$) such that $\alpha'_i Q = \beta'_i$ for $i = 1, \dots, n$, which gives

$$\phi_Q(A_i) = \frac{\sqrt{\alpha_i \alpha_i^T}}{\sqrt{\beta_i \beta_i^T}} \gamma_i^T \beta_i, \quad i = 1, 2, \dots, n.$$

As $\frac{\sqrt{\alpha_i \alpha_i^T}}{\sqrt{\beta_i \beta_i^T}} \gamma_i^T$ and λ_i^T are nonzero, there is $P_i \in G$ such that $P_i(\frac{\sqrt{\alpha_i \alpha_i^T}}{\sqrt{\beta_i \beta_i^T}} \gamma_i^T) = \lambda_i^T$. Thus $\phi_Q(A_i) \stackrel{G}{\sim} B_i$ for $i = 1, 2, \dots, n$, which proves the first assertion.

Let ρ be the mapping which sends $\phi_Q(A_i)$ to $P_i \cdot \phi_Q(A_i)$ and sends B_i to $P_i^{-1} B_i$ for $i = 1, 2, \dots, n$ and it fixes all other vertices. Then ρ is an automorphism of the graph and the automorphism $\rho \cdot \phi_Q$ sends A_i to B_i for $i = 1, 2, \dots, n$. $\square$

3. Proof of the main results

Now, we consider the action of automorphisms on the set of rank one matrices, which is key for determining the automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$. Let V_1 be the set of rank one matrices in $M_{m,n}(\mathbb{F})$.

Lemma 3.1. *The set V_1 is stable under the action of automorphisms of $\Omega(M_{m,n}(\mathbb{F}))$ and the action of automorphisms on V_1 is transitive.*

Proof. Let $A \in V_1$ be a rank one matrix and let σ be an automorphism of $\Omega(M_{m,n}(\mathbb{F}))$. By Lemma 2.3, A lies in a maximum clique $\mathcal{M}$. Thus $\sigma(A)$ lies in the maximum clique $\sigma(\mathcal{M})$. By Lemma 2.2, $\sigma(A)$ is also a rank one matrix.

For any two rank one matrices A and B , by Lemma 2.3, we may suppose A and B respectively lie in maximum cliques $\mathcal{A}$ and $\mathcal{B}$. By Lemma 2.4, there is an automorphism sends A to B . Thus the action of automorphisms on V_1 is transitive. $\square$

If only the first row of a rank one matrix A is nonzero, it can be written as

$$A = a_1 E_{1,1} + \dots + a_n E_{1,n}, \quad a_i \in \mathbb{F}.$$

Such a rank one matrix is called *standard*. Furthermore, let $A = a_1 E_{1,1} + \dots + a_n E_{1,n}$ be a standard rank one matrix, if the first nonzero element among $a_1, a_2, \dots, a_n$ is 1, we call A a *monic standard* rank one matrix. Indeed, any rank one matrix in $M_{m,n}(\mathbb{F})$ can be decomposed into the product of an invertible matrix in G and a monic standard rank one matrix.

Lemma 3.2. *For a rank one matrix $A \in M_{m,n}(\mathbb{F})$, there is a unique monic standard rank one matrix A_0 such that $A \stackrel{G}{\sim} A_0$.*

Proof. Since the rank of A is 1, there are nonzero vectors $\alpha \in \mathbb{F}^n, \gamma \in \mathbb{F}^m$ such that $A = \gamma^T \alpha$. Assume that $\alpha = a_1 e_1 + \dots + a_n e_n$ and assume that $a_1 = \dots = a_{i-1} = 0, a_i \neq 0$ with $1 \leq i \leq n$. There is $P \in G$ such that $P\gamma^T = e_1^T$. Thus $PA = a_1 E_{1,1} + \dots + a_n E_{1,n}$, which follows $A = P_0 A_0$, where $A_0 = \frac{a_1}{a_i} E_{1,1} + \dots + \frac{a_j}{a_i} E_{1,j} + \dots + \frac{a_n}{a_i} E_{1,n}$ is monic standard and $P_0 = a_i P^{-1} \in G$.

Let $A = P'A'$ be another decomposition of A with $P' \in G$ and $A' = a'_1 E_{1,1} + \dots + a'_n E_{1,n}$ is monic standard. Suppose the k th row of A is nonzero. Clearly, the k th row of $P'A'$ is $b'a'_1 e_1 + \dots + b'a'_n e_n$ and the k th row of $P_0 A_0$ is $\frac{b_0 a_1}{a_i} e_1 + \dots + \frac{b_0 a_j}{a_i} e_j + \dots + \frac{b_0 a_n}{a_i} e_n$, where b' and b_0 are respectively the $(k, 1)$ -entry of P' and P_0 . Clearly, b', b_0 are nonzero. The monic condition for A' and A_0 implies $b' = b_0$. Hence, $a'_j = \frac{a_j}{a_i}$ for $j = 1, \dots, n$ and thus $A' = A_0$. $\square$

For two vertices A, B of $\Omega(M_{m,n}(\mathbb{F}))$, we say A and B have the *same neighbours* if $A \sim C$ if and only if $B \sim C$. Now, we consider when A and B have the same neighbours.

Lemma 3.3. *Two vertices A and B of $\Omega(M_{m,n}(\mathbb{F}))$ have the same neighbours if and only if A and B are row-equivalent.*

Proof. Suppose $B = PA$ with $P \in G$. If $A \sim C$ for a vertex C of $\Omega(M_{m,n}(\mathbb{F}))$, then $A\bar{C}^T = 0$, which follows that $B\bar{C}^T = 0$ and $B \sim C$. Conversely, if $B \sim C$, then $B\bar{C}^T = 0$, which follows $A\bar{C}^T = (P^{-1}B)\bar{C}^T = 0$, showing that $A \sim C$. Hence, A and B have the same neighbours.

Suppose A and B have the same neighbours, we need firstly prove that the linear equations $AX = 0$ and $BX = 0$ have the same solutions. Indeed, if α^T with $\alpha \in \mathbb{F}^n$ is a nonzero solution of $AX = 0$, that is $A\alpha^T = 0$. Let $C_i = e_i^T \bar{\alpha}$ for $i = 1, 2, \dots, n$. Then from $A \sim C_i$ it follows that $B \sim C_i$ for $i = 1, 2, \dots, n$ and thus $B\alpha^T = 0$. Similarly, any solution of $BX = 0$ is that of $AX = 0$. Hence, $\begin{pmatrix} A \\ B \end{pmatrix}$ and A have the same rank, from which we know that the row vectors of B can be written as the linear combinations of the row vectors of A . Hence, $B = PA$ for some $P \in G$. $\square$

Now, we are ready to give a proof for the main results.

Proof of Theorems 1.1 and 1.2. The sufficiency is obvious, we only consider the necessity. Let σ be an automorphism of $\Omega(M_{m,n}(\mathbb{F}))$. By Lemma 2.1, the set $\{E_{1,1}, E_{1,2}, \dots, E_{1,n}\}$ induces a maximum clique of $\Omega(M_{m,n}(\mathbb{F}))$. Thus $\{\sigma(E_{1,1}), \sigma(E_{1,2}), \dots, \sigma(E_{1,n})\}$ also induces a maximum clique of the graph. By applying Lemma 2.4, there is a unitary matrix Q (or an orthogonal matrix if $\mathbb{F}$ is the real field) such that

$$\sigma_1(E_{1,j}) \stackrel{G}{=} E_{1,j}, \quad j = 1, 2, \dots, n,$$

where $\sigma_1 = \phi_Q \cdot \sigma$ is an automorphism of $\Omega(M_{m,n}(\mathbb{F}))$.

For $1 \leq i < j \leq n$ and $a \in \mathbb{F}$, we consider the action of σ_1 on $E_{1,i} + aE_{1,j}$. By Lemmas 3.1 and 3.2, we may assume that

$$\sigma_1(E_{1,i} + aE_{1,j}) \stackrel{G}{=} b_1E_{1,1} + \dots + b_nE_{1,n},$$

where $b_1E_{1,1} + \dots + b_nE_{1,n}$ is monic standard. Applying σ_1 to $(E_{1,i} + aE_{1,j}) \sim E_{1,k}$ for $k \notin \{i, j\}$, we have $\sigma_1(E_{1,i} + aE_{1,j}) \sim E_{1,k}$, which implies that $b_k = 0$ for $k \notin \{i, j\}$. Thus

$$\sigma_1(E_{1,i} + aE_{1,j}) \stackrel{G}{=} E_{1,i} + b_jE_{1,j},$$

and $b_j = 0$ if and only if $a = 0$. Now, σ_1 induces a function f_{ij} on $\mathbb{F}$ such that

$$\sigma_1(E_{1,i} + aE_{1,j}) \stackrel{G}{=} E_{1,i} + f_{ij}(a)E_{1,j}, \quad \forall a \in \mathbb{F}.$$

We will give nine claims to reveal some properties of f_{ij} . It has been shown that $f_{ij}(0) = 0$. For $a \neq 0$, by applying σ_1 to

$$(E_{1,i} + aE_{1,j}) \sim (E_{1,i} - \bar{a}^{-1}E_{1,j}),$$

we have

$$(E_{1,i} + f_{ij}(a)E_{1,j}) \sim (E_{1,i} + f_{ij}(-\bar{a}^{-1})E_{1,j}),$$

and thus we have:

Claim 1. $f_{ij}(a) = -\overline{f_{ij}(-\bar{a}^{-1})}^{-1}$ for $1 \leq i < j \leq n$ and $0 \neq a \in \mathbb{F}$,

Now, we consider the action of σ_1 on $E_{1,i} + \sum_{j=i+1}^n a_jE_{1,j}$. Suppose

$$\sigma_1(E_{1,i} + \sum_{j=i+1}^n a_jE_{1,j}) \stackrel{G}{=} \sum_{k=1}^n b_kE_{1,k}.$$

Noting that $E_{1,i} + \sum_{j=i+1}^n a_jE_{1,j} \sim E_{1,k}$ for $k < i$, thus we have $b_k = 0$ for $k < i$. Similarly, $b_j = 0$ if and only if $a_j = 0$ for $j > i$. As $b_i \neq 0$, we may assume $b_i = 1$. For $i < j \leq n$, if $a_j \neq 0$, the action of σ_1 to

$$(E_{1,i} + \sum_{j=i+1}^n a_jE_{1,j}) \sim (E_{1,i} - \bar{a}_j^{-1}E_{1,j}),$$

derives that

$$(E_{1,i} + \sum_{j=i+1}^n b_jE_{1,j}) \sim (E_{1,i} + f_{ij}(-\bar{a}_j^{-1})E_{1,j}),$$

which follows that $1 + b_j \cdot \overline{f_{ij}(-\bar{a}_j^{-1})} = 0$. Applying Claim 1, we have $b_j = f_{ij}(a_j)$ for $j = i+1, \dots, n$. Thus we have:

Claim 2. $\sigma_1(E_{1,i} + \sum_{j=i+1}^n a_jE_{1,j}) \stackrel{G}{=} E_{1,i} + \sum_{j=i+1}^n f_{ij}(a_j)E_{1,j}$ for $1 \leq i \leq n-1$.

Now, we study the relation between the functions f_{ij} . For nonzero $a, b \in \mathbb{F}$ and $1 < i < j \leq n$, the action of σ_1 to

$$(E_{1,1} + aE_{1,i} + abE_{1,j}) \sim (E_{1,i} - \bar{b}^{-1}E_{1,j}),$$

derives that

$$(E_{1,1} + f_{1i}(a)E_{1,i} + f_{1j}(ab)E_{1,j}) \sim (E_{1,i} + f_{ij}(-\bar{b}^{-1})E_{1,j}),$$

which follows that $f_{1i}(a) + f_{1j}(ab)\overline{f_{ij}(-\bar{b}^{-1})} = 0$. Thus $f_{1j}(ab) = -\overline{f_{1i}(a)f_{ij}(-\bar{b}^{-1})}^{-1}$. By claim 1, we have $f_{1j}(ab) = f_{1i}(a)f_{ij}(b)$. If $ab = 0$, this formula holds naturally. Thus we have:

Claim 3. $f_{1j}(ab) = f_{1i}(a)f_{ij}(b)$ for $1 < i < j \leq n$ and $a, b \in \mathbb{F}$.

Taking $i = 2$ and $b = 1$ or $a = 1$ in Claim 3, we have $f_{12}(a) = f_{1j}(a)f_{2j}(1)^{-1}$ and $f_{2j}(b) = f_{1j}(b)f_{12}(1)^{-1}$. By substituting them into $f_{1j}(ab) = f_{12}(a)f_{2j}(b)$ we have

$$f_{1j}(ab) = f_{1j}(a)f_{1j}(b)f_{1j}(1)^{-1} \quad \text{for } 2 < j \leq n, \quad a, b \in \mathbb{F}.$$

The case when $j = 2$ also needs be considered. We have obtained $f_{12}(a) = f_{1j}(a)f_{2j}(1)^{-1}$ for $2 < j \leq n$, replacing a with ab , we have $f_{12}(ab) = f_{1j}(ab)f_{2j}(1)^{-1}$. Substituting $f_{1j}(ab) = f_{1j}(a)f_{1j}(b)f_{1j}(1)^{-1}$ into it, we have

$$f_{12}(ab) = f_{1j}(a)f_{1j}(b)f_{1j}(1)^{-1}f_{2j}(1)^{-1}.$$

Recalling that $f_{1j}(a) = f_{12}(a)f_{2j}(1)$, $f_{1j}(b) = f_{12}(b)f_{2j}(1)$ and $f_{1j}(1) = f_{12}(1)f_{2j}(1)$, we have

$$f_{12}(ab) = f_{12}(a)f_{12}(b)f_{12}^{-1}(1).$$

Combining two cases together, we have:

Claim 4. $f_{1j}(ab) = f_{1j}(a)f_{1j}(b)f_{1j}(1)^{-1}$ for $2 \leq j \leq n$ and $a, b \in \mathbb{F}$.

When $a \neq 0$, by taking $b = a^{-1}$, the equality in Claim 4 gives the following claim.

Claim 5. $f_{1j}(a^{-1}) = f_{1j}(a)^{-1}f_{1j}(1)^2$ for $a \neq 0$ and $2 \leq j \leq n$.

Let $f = f_{12}(1)^{-1} \cdot f_{12}$. Thus $f(1) = 1$. For $a, b \in \mathbb{F}$, $f(ab) = f_{12}(ab)f_{12}(1)^{-1}$. Substituting $f_{12}(ab) = f_{12}(a)f_{12}(b)f_{12}(1)^{-1}$ into it, we have $f(ab) = f(a)f(b)$. Taking $a = b = -1$, we have $1 = f(1) = f(-1)^2$. Since $f_{12}(-1) \neq f_{12}(1)$, we have $f(-1) \neq 1$ and thus $f(-1) = -1$. Thus we have the productive property of f :

Claim 6. $f(ab) = f(a)f(b)$ for $a, b \in \mathbb{F}$, $f(1) = 1$ and $f(-1) = -1$.

Now, we study the additive property of f . If $a = 0$, then $f_{1j}(a + b) = f_{1j}(a) + f_{1j}(b)$ for $2 \leq j \leq n$. Now we consider the case when $a \neq 0$. Applying σ_1 to

$$(E_{1,1} + bE_{1,2} + (a + b)E_{1,j}) \sim (E_{1,1} + \bar{a}^{-1}E_{1,2} - \bar{a}^{-1}E_{1,j}), \quad j \geq 3,$$

we have

$$(E_{1,1} + f_{12}(b)E_{1,2} + f_{1j}(a + b)E_{1,j}) \sim (E_{1,1} + f_{12}(\bar{a}^{-1})E_{1,2} + f_{1j}(-\bar{a}^{-1})E_{1,j}),$$

which follows that

$$1 + f_{12}(b)f_{12}(\bar{a}^{-1}) + f_{1j}(a + b)f_{1j}(-\bar{a}^{-1}) = 0.$$

Since $\overline{f_{1j}(-\bar{a}^{-1})} = -f_{1j}(a)^{-1}$ and $\overline{f_{12}(\bar{a}^{-1})} = -f_{12}(-a)^{-1}$ (by Claim 1), we further have

$$f_{1j}(a + b) = f_{1j}(a) - f_{1j}(a)f_{12}(b)f_{12}(-a)^{-1}.$$

Claim 3 implies that $f_{1j}(a)f_{12}(b) = f_{1j}(b)f_{12}(a)$. Substituting it to the above equality, we have

$$f_{1j}(a + b) = f_{1j}(a) - f_{1j}(b)f_{12}(a)f_{12}(-a)^{-1}.$$

Claim 3 says $f_{12}(a) = f_{12}(-1)f_{12}(-a)f_{12}(1)^{-1}$. Substituting it to the above equality, we have

$$f_{1j}(a + b) = f_{1j}(a) - f_{1j}(b)f_{12}(-1)f_{12}(1)^{-1}.$$

As $f_{12}(-1)f_{12}(1)^{-1} = f(-1) = -1$, we arrive at

$$f_{1j}(a + b) = f_{1j}(a) + f_{1j}(b), \quad \text{for } 3 \leq j \leq n, \quad a, b \in \mathbb{F}.$$

For the case when $j = 2$, by substituting $f_{13}(x) = f_{12}(x)f_{23}(1)$ with $x = a + b$, a or b to $f_{13}(a + b) = f_{13}(a) + f_{13}(b)$, we have

$$f_{12}(a + b) = f_{12}(a) + f_{12}(b), \quad \text{for } a, b \in \mathbb{F}.$$

By the definition of f , we establish the additive property of f as follows.

Claim 7. $f(a + b) = f(a) + f(b)$ for $a, b \in \mathbb{F}$.

Clearly, f is a bijective mapping on $\mathbb{F}$. Combining this with Claim 6 and Claim 7, we know f is a field automorphism of $\mathbb{F}$. The automorphism of $\mathbb{C}$ or $\mathbb{R}$ is well known. Indeed, $\mathbb{R}$ has only the identity automorphism and $\mathbb{C}$ has a unique nontrivial automorphism, which is the mapping sending any $a \in \mathbb{C}$ to its conjugate $\bar{a}$. Thus we have:

Claim 8. Either $f(a) = a$ or $f(a) = \bar{a}$ for $a \in \mathbb{F}$.

Now, we study the relation between f_{ij} and f . For $j \geq 3$, $a \in \mathbb{F}$, by $f_{1j}(a) = f_{12}(a)f_{2j}(1)$ and $f_{1j}(1) = f_{12}(1)f_{2j}(1)$ we have

$$f_{1j}(a)f_{1j}(1)^{-1} = f_{12}(a)f_{12}(1)^{-1} = f(a).$$

Thus we have: $f_{1j}(a) = f_{1j}(1)f(a)$ for $2 \leq j \leq n$ and $a \in \mathbb{F}$. Next, for $2 \leq i < j \leq n$ and $a \in \mathbb{F}$, we have

$$f_{ij}(a) = f_{ii}(1)^{-1}f_{1j}(a) = f_{ii}(1)^{-1}f_{1j}(1)f(a) = f_{ij}(1)f(a).$$

Set $f_{11}(1) = 1$. Thus, we have

Claim 9. $f_{ij} = f_{ij}(1)f = f_{1i}(1)^{-1}f_{1j}(1)f$ for $1 \leq i < j \leq n$.

By Claim 1, $f_{ij}(1)\overline{f_{ij}(-1)} = -1$ and by Claim 9, $f_{ij}(-1) = f_{ij}(1)f(-1) = -f_{ij}(1)$. Thus, $f_{ij}(1)\overline{f_{ij}(1)} = 1$ for $j = 1, \dots, n$. Let $D = \text{diag}(f_{11}(1), f_{12}(1), \dots, f_{1n}(1))$. Then by the above equalities, we have $DD^T = I$, showing that D is a unitary matrix. By Claim 2 and Claim 9,

$$\sigma_1(E_{1,i} + \sum_{j=i+1}^n a_j E_{1,j}) \stackrel{G}{=} E_{1,i} + \sum_{j=i+1}^n f_{1i}(1)^{-1}f_{1j}(1)f(a_j)E_{1,j}, \quad 1 \leq i \leq n.$$

Thus

$$(\phi_D^{-1} \cdot \sigma_1)(E_{1,i} + \sum_{j=i+1}^n a_j E_{1,j}) \stackrel{G}{=} E_{1,i} + \sum_{j=i+1}^n f(a_j)E_{1,j}, \quad 1 \leq i \leq n.$$

The extension of f to $M_{m,n}(\mathbb{F})$ is also written as f . Let $\sigma_2 = f^{-1} \cdot \phi_D^{-1} \cdot \sigma_1$. Then

$$\sigma_2(E_{1,i} + \sum_{j=i+1}^n a_j E_{1,j}) \stackrel{G}{=} E_{1,i} + \sum_{j=i+1}^n a_j E_{1,j}, \quad 1 \leq i \leq n.$$

This is to say that σ_2 sends every monic standard rank one matrix $A = E_{1,i} + \sum_{j=i+1}^n a_j E_{1,j}$ to $P_A A$ with $P_A \in G$, where $P_A A$ and A have the same neighbours.

Next, we consider the action of σ_2 on any given vertex of $\mathcal{Q}(M_{m,n}(\mathbb{F}))$. For a rank one matrix A , it can be written as the product of an invertible matrix and a monic standard rank one matrix as $A = CA_0$ with $C \in G$ and A_0 monic standard. Thus A and A_0 have the same neighbours. Since σ_2 sends A_0 to a rank one matrix with the same neighbours, σ_2 sends A to a rank one matrix with the same neighbours as A . That is

Claim 10. $\sigma_2(A) = P_A A$ for any rank one matrix A , where $P_A \in G$ depends on A .

Now, we consider the action of σ_2 on any nonzero matrix A with rank less than n . We will prove that $\sigma_2(A)$ and A have the same neighbours. Suppose $A \sim B$ and suppose β is a nonzero row vector of B . It follows from $A\bar{\beta}^T = 0$ that $A\bar{\beta}^T = 0$. Thus we have $A \sim e_i^T \beta$ and further $\sigma_2(A) \sim \sigma_2(e_i^T \beta)$ for $i = 1, 2, \dots, m$. Since $\sigma_2(e_i^T \beta) = C_i(e_i^T \beta)$ for some $C_i \in G$, it follows that $\sigma_2(A) \cdot (\beta^T e_i C_i^T) = 0$ for $i = 1, 2, \dots, m$. Hence, $\sigma_2(A) \cdot \bar{\beta}^T = 0$ and further $\sigma_2(A) \cdot \bar{B}^T = 0$, which leads to $\sigma_2(A) \sim B$. By considering the inverse of σ_2 , we have $A \sim B$ if $\sigma_2(A) \sim B$. Thus $\sigma_2(A)$ and A have the same neighbours. It proves the following claim.

Claim 11. $\sigma_2(A) = P_A A$ for any nonzero matrix $A \in M_{m,n}(\mathbb{F})$ with rank less than n , where $P_A \in G$ depends on A .

Back to σ , we have $\sigma(A) = f(P_A)f(A)DQ^{-1}$. Denote $f(P_A)$ by X_A , which also lies in G and depends on A , and denote DQ^{-1} by U , which is also a unitary matrix (or an orthogonal matrix). Finally, we have $\sigma(A) = X_A f(A)U$ for any nonzero $A \in M_{m,n}(\mathbb{F})$ with rank less than n . The proof is completed. $\square$

For the special case when $n = 2$, observe that any vertex A in $\mathcal{Q}(M_{m,2}(\mathbb{F}))$ is a rank one matrix. If a rank one matrix A for which $Ae_1^T \neq 0$ (the first column of A is nonzero), it can be uniquely written as $A = \begin{pmatrix} a_1 & a_1 c \\ a_2 & a_2 c \\ \dots & \dots \\ a_m & a_m c \end{pmatrix}$. In this case, a

new type of automorphism emerges.

• Let τ be a bijective mapping on $\mathbb{F}$ such that $\tau(0) = 0$ and $\tau(-\bar{a}^{-1}) = -\tau(a)^{-1}$. Then the mapping on the vertex set of $\mathcal{Q}(M_{m,2}(\mathbb{F}))$, which sends $A = \begin{pmatrix} a_1 & a_1 c \\ a_2 & a_2 c \\ \dots & \dots \\ a_m & a_m c \end{pmatrix}$ to $A = \begin{pmatrix} a_1 & a_1 \tau(c) \\ a_2 & a_2 \tau(c) \\ \dots & \dots \\ a_m & a_m \tau(c) \end{pmatrix}$ and which fixes all rank one matrix A for which $Ae_1^T = 0$. It is easy to check such a mapping is an automorphism of $\mathcal{Q}(M_{m,2}(\mathbb{F}))$. The action of such an automorphism on a vertex A of $\mathcal{Q}(M_{m,2}(\mathbb{F}))$ is written as A^τ .

Theorem 3.4. A bijective mapping φ on the vertex set of $\mathcal{Q}(M_{m,2}(\mathbb{F}))$ is an automorphism if and only if there is a permutation τ on $\mathbb{F}$ satisfying $\tau(0) = 0$ and $\tau(-\bar{a}^{-1}) = -\tau(a)^{-1}$ such that

$$\varphi(A) = PA^\tau Q,$$

where $P \in M_m(\mathbb{F})$ is invertible and $Q \in M_n(\mathbb{F})$ satisfies $Q\bar{Q}^T = I$, and $A^\tau = A$ if $Ae_1^T = 0$ and $A^\tau = \begin{pmatrix} a_1 & a_1\tau(c) \\ a_2 & a_2\tau(c) \\ \dots & \dots \\ a_m & a_m\tau(c) \end{pmatrix}$ if

$$A = \begin{pmatrix} a_1 & a_1c \\ a_2 & a_2c \\ \dots & \dots \\ a_m & a_mc \end{pmatrix} \text{ such that } Ae_1^T \neq 0.$$

Proof. The sufficiency is obvious. Let σ be an automorphism of $\Omega(M_{m,2}(\mathbb{F}))$. As the proof for Theorem 1.1 and 1.2, the set $\{E_{1,1}, E_{1,2}\}$ induces a maximum clique of $\Omega(M_{m,2}(\mathbb{F}))$. Thus $\{\sigma(E_{1,1}), \sigma(E_{1,2})\}$ also induces a maximum clique of the graph. By applying Lemma 2.4, there is a unitary matrix Q (or an orthogonal matrix if $\mathbb{F}$ is the real field) such that $(\phi_Q \cdot \sigma)(E_{1,j}) \stackrel{G}{=} E_{1,j}$, $j = 1, 2$. Let $\sigma_1 = \phi_Q \cdot \sigma$. We have $\sigma_1(E_{1,i}) \stackrel{G}{=} E_{1,i}$, $i = 1, 2$. The action of σ_1 on $E_{1,1} + cE_{1,2}$ induces a function τ on $\mathbb{F}$ in such a way:

$$\sigma_1(E_{1,1} + cE_{1,2}) \stackrel{G}{=} E_{1,1} + \tau(c)E_{1,2}.$$

Obviously, $\tau(0) = 0$. For $0 \neq c \in \mathbb{F}$, by applying σ_1 to

$$E_{1,1} + cE_{1,2} \sim E_{1,1} - \bar{c}^{-1}E_{1,2},$$

we have

$$E_{1,1} + \tau(c)E_{1,2} \sim E_{1,1} + \tau(-\bar{c}^{-1})E_{1,2},$$

which leads to

$$\tau(-\bar{c}^{-1}) = -\tau(c)^{-1}, \quad \forall 0 \neq c \in \mathbb{F}.$$

Now, let $\sigma_2 = \tau^{-1} \cdot \sigma_1$. Then σ_2 sends every $E_{1,1} + cE_{1,2}$ with $c \in \mathbb{F}$ to a vertex with the same neighbours. For

$A = \begin{pmatrix} a_1 & a_1c \\ a_2 & a_2c \\ \dots & \dots \\ a_m & a_mc \end{pmatrix}$, there is an invertible 2×2 matrix P such that $A = P(E_{1,1} + cE_{1,2})$, which has the same neighbours

as that of $E_{1,1} + cE_{1,2}$. As σ_2 sends $E_{1,1} + cE_{1,2}$ to a vertex which has the same neighbours, σ_2 sends A to a vertex which has the same neighbours as that of A . Hence, $\sigma_2(A) = P_A A$ with $P_A \in G$. If Ae_1^T is zero, similar discussion can also prove that $\sigma_2(A) = P_A A$ with $P_A \in G$. Finally, as the proof for Theorem 1.1, 1.2, we have $\sigma(A) = X_A A^\tau U$ for any rank one matrix $A \in M_{m,2}(\mathbb{F})$, where $X_A \in G$ and U is a unitary matrix (or an orthogonal matrix). $\square$

Acknowledgements

We gratefully acknowledge the support of the Graduate Innovation Program of China university of Mining and Technology (2024WLKXJ119) and the Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX24_2678) for this work. We appreciate the detailed comments given by the referee, and we believe these comments can improve the clarity and quality of our paper.

Data availability

No data was used for the research described in the article.

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